

Research-Informed Mathematics Teaching Design

A district-facing mathematics workspace for stronger teaching, faster planning, and more consistent classroom practice.

Kaizen Maths helps teachers plan faster, teach with clearer worked examples, generate high-quality practice, create assessments, and use classroom-ready representations from one professional mathematics workspace. Its design is grounded in established principles from mathematics education, learning science, formative assessment, and purposeful educational technology.

District value

Kaizen Maths strengthens implementation by giving every teacher access to the same structured question bank, worked examples, curriculum maps, classroom displays, worksheets, assessments, and pupil practice routines.

Why this matters for districts and schools

Save planning time Move from topic selection to board practice, worksheet, quiz, or assessment in minutes.	Improve consistency Give teachers shared topic structures, question levels, curriculum links, and worked-solution routines.	Support intervention Generate focused practice for prerequisites, revision, catch-up, and targeted class needs.	Scale routines Use training, tool guidance, misconceptions, and classroom questions to support implementation.
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Research-supported design principles inside Kaizen Maths

Mastery progression Topic levels and mixed sets help teachers sequence practice coherently.	Retrieval and fluency Fresh generated questions support repeated recall and deliberate practice.	Worked examples Step reveal and one-example mode make methods visible for teachers and pupils.
Representations Bar models, area models, diagrams, displays, and dynamic tools expose structure.	Feedback Answers, worked steps, pupil tasks, and review tools support responsive teaching.	Teacher-led technology The teacher controls the task, pace, reveal points, and classroom routine.
Virtual textbook Topic-based question generation, levels, worked examples, worksheets, assessments, and displays.	Classroom instruction Projection, step reveal, annotation, capture, and dynamic representations support live teaching.	Assessment and feedback Pupil tasks, mastery thresholds, review comments, answer keys, and worked steps make practice actionable.

Teacher wellbeing and workload

When implemented effectively, Kaizen reduces planning pressure and decision fatigue. Teachers spend less time searching for questions, rebuilding worksheets, formatting assessments, locating diagrams, and preparing worked examples. That time can move back into explanation, feedback, relationships, and responsive teaching.

Selected research links

EEF mathematics evidence review educationendowmentfoundation.org.uk/.../mathematics-in-key-stages-2-and-3	WWC mathematics intervention guide ies.ed.gov/ncee/WWC/PracticeGuide/26
WWC algebra practice guide ies.ed.gov/ncee/WWC/PracticeGuide/20	EEF feedback guidance educationendowmentfoundation.org.uk/.../feedback
Dunlosky et al. learning techniques psychologicalscience.org/publications/journals/pspi/learning-techniques.html	Roediger and Karpicke retrieval practice doi.org/10.1111/j.1467-9280.2006.01693.x
NCETM five big ideas in mastery ncetm.org.uk/teaching-for-mastery	EEF digital technology guidance educationendowmentfoundation.org.uk/.../digital